

Muscle Tissue Practice Supplement — ANSWER KEY

TOPIC: STRUCTURE OF A SKELETAL MUSCLE

The Muscle Fiber

- For a muscle fiber to contract, the tissue must **[be stimulated / receive a signal]**.
- Sarco**[lemma]**: plasma membrane wrapping myo**[fibrils]**
 - **T-Tubules**: membrane **[invaginations / extensions]** that reach deep myofibrils.
- Myo**[fibrils]**: long, rod-shaped organelles.
 - Contain myo**[filaments]**.
- Myo**[filaments]**: Proteins: actin & myosin.
- Sarco**[mere]**: Contractile **[unit]**.
 - Creates **[striated]** appearance.
- Sarco[plasmic] Reticulum**: Endoplasmic reticulum of muscle cells.
 - Surrounds myofibrils.
 - Store/release Ca^{2+} ions.

TOPIC: SLIDING FILAMENT THEORY AND THE SARCOMERE

Sliding Filament Theory

- How do muscles get shorter?
 - **Myosin** (**[thick]** filament): anchored to the *center* of the sarcomere.
– *Protein that wants to pull on a rope.*
 - **Actin** (**[thin]** filament): anchored to the *ends* of the sarcomere.
– *Protein that is the [rope].*
- During contraction:
 - Filaments (actin & myosin) stay the **[same]** length; overlap of actin & myosin **[increases]**.

EXAMPLE — Sarcomere Labeling

1. Myosin filaments: the **thick** filaments in the center of each sarcomere (between Z-discs).
2. Actin filaments: the **thin** filaments extending from the Z-discs toward the center.
3. When the muscle contracts, the **[actin]** filament gets closer to the center of the sarcomere.

TOPIC: SLIDING FILAMENT THEORY AND THE SARCOMERE

Proteins of the Sarcomere

- *Recall*: Sarcomere is the contractile unit: **[myosin]** pulls on **[actin]**.
- **Contractile Proteins**:
 - **Myosin**: **[thick]** filament: *many headed Medusa.*
 - **Actin**: **[thin]** filament: *acTIN is THIN.*
- **Regulatory Proteins**:
 - **Tropomyosin**: **[rod]**-like protein, wraps actin.
– **[Blocks / Covers]** the myosin binding sites on actin.
 - **Troponin**: **[regulatory]** protein.
– *Binds to Ca^{2+} : **[exposes / uncovers]** the binding sites on actin by moving tropomyosin.*
– *Tropomyosin says "No" to the myosin. TROPONin OPENS the binding site.*
- **Structural Proteins**:

- **Elastic Filament** (Titin): helps sarcomere retain **[shape / elasticity]**.

Structure of the Sarcomere: Bands, Zones, Discs & Lines

- The regions of the sarcomere were named for how they **[appeared / looked]** on a TEM microscope.

1. Discs and Lines:

- **[Z]** Disc: end of sarcomere. (*Z is at the end of the alphabet.*)
- **[M]** Line: Myomesin protein that **[anchors]** myosin. (*M = Middle.*)

2. Bands and Zones:

- **I band** (**[light]** Band): area with just **[actin]**.
- **A band** (**[dark]** Band): area with **[both]** actin & myosin.
- **[H]** Zone: center region with just **[myosin]**. (*Helle = bright.*)

TOPIC: STEPS OF MUSCLE CONTRACTION

EXAMPLE — Sarcomere Regions During Contraction

- a) Mark A Band, I Band, H Zone with brackets on the TEM image. b) Mark Z Disc and M Line with arrows. c) Draw a line = length of one Actin filament.

Structure	Description	Change in size during contraction
A Band	Region containing myosin (thick filaments); overlaps with actin	[No change — stays the same size]
I Band	Region with only actin (thin filaments), flanking the Z-discs	[Decreases / gets smaller]
H Zone	Center of A band; only myosin, no actin overlap	[Decreases / gets smaller]
Z Disc	Boundary/end of each sarcomere; actin anchored here	[Move closer together]
M Line	Center of sarcomere; myosin anchored here	[No change — stays the same]

Overview of Muscle Contraction

- Muscle contraction involves the **[transmission of a signal via the]** nervous **[system]**, initiating an action potential.
- A.** At the neuro-muscular junction, muscle cell is stimulated by the **[nervous]** system, initiating an action potential.
- B. Excitation-Contraction Coupling:**
 - Action potential **[propagates]** along the sarcolemma and enters the **[T]**-tubules.
 - Sarcoplasmic reticulum releases **[Ca²⁺]**; **[myosin]** binding sites are exposed.
- C.** Myosin binds to **[actin]**, forming a **[cross]**bridge and performs the **power stroke**.

TOPIC: STEPS OF MUSCLE CONTRACTION

Neurotransmitters & Action Potentials

EXAMPLE — Classify each structure (T = transfers signal, R = regulates contraction, C = directly active in mechanics):

Structure	Answer	Reason
1. Actin	[C]	Directly performs contraction (thin filament pulled by myosin)
2. Calcium ions	[R]	Binds troponin to regulate (expose) myosin binding sites
3. Myosin	[C]	Directly performs contraction (power stroke)
4. Sarcolemma	[T]	Transfers action potential along muscle cell surface

5. Sarcoplasmic reticulum	[T]	Transfers/delivers Ca^{2+} signal into the cell interior
6. Troponin	[R]	Regulatory protein — opens binding sites when Ca^{2+} binds
7. Tropomyosin	[R]	Regulatory protein — blocks/unblocks actin binding sites
8. T tubule	[T]	Transfers action potential deep into the muscle fiber

Neurotransmitters & Action Potentials — Filled Blanks

- **Synapse:** [gap / space] between the axon and the muscle.
- **Acetylcholine:** neurotransmitter of the neuromuscular junction.
- **Neurotransmitters:** chemical messengers used at [synapses / the NMJ].

Action Potential — Depolarization & Repolarization:

- Change of polarization caused by movement of Na^{+} and K^{+} .

State	Ion movement	Result
Polarized (resting)	Na^{+} [outside] cell; K^{+} [inside] cell	Outside positive, inside negative
Depolarization	Na^{+} [moves inside]	Inside cell becomes more positive
Repolarization	K^{+} [moves outside]	Charge is restored (resting state returns)

All answers shown in red. Original worksheet: Muscle Tissue Practice Supplement.